QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 YEAR ADMISSION ONWARDS)

MG4DSECFS200 - VIRTUAL PRIVATE NETWORK SECURITY

(Network Security Specialization)

Duration: 2hrs **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Identify the main purpose of a Virtual Private Network (VPN). [Understand] [CO:1]
2. Explain why encryption is important in a VPN. [Understand] [CO:1,2]
3. Define the term Site-to-Site VPN. [Understand] [CO:1]
4. Simplify the use of one VPN protocol for securing data communication [Analyze] [CO:2]
5. Write about one encryption algorithm used in VPNs. [Analyze] [CO:2,3]
6. Simplify the term L2TP. [Analyze] [CO:2]
7. Identify one common user authentication method. [Understand] [CO:1]
8. Explain the meaning of RBAC in managing VPN users. [Apply] [CO:3]

9.	Outline the purpose of LDAP.	[Analyze] [CO:2,3]
10.	Explain the function of a digital certificate in VPN authentication.	[Understand] [CO:1,4]
11.	Write about the VPN deployment method used to connect two branch offices.	[Apply] [CO:3]
12.	Identify one method to improve the performance of a VPN.	[Analyze] [CO:2]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

13.	Explain the different techniques used for eavesdropping attacks on VPN communications.	[Understand][CO:1,2]
14.	Differentiate between DES and AES with respect to key size, block size, security level, and performance.	[Analyze][CO:2,3]
15.	Compare LDAP and RADIUS for VPN integration and explain how Network Access Control (NAC) helps in managing user access.	[Analyze][CO:2,3]
16.	Illustrate the strength of Two-Factor Authentication (2FA) compared to single-factor authentication using a suitable real-world scenario.	[Evaluate][CO:1,4]
17.	Write about the role of RADIUS in VPN authentication and its benefits for security and user management.	[Apply][CO:3]
18.	Describe the role of regular updates, patch management, and continuous monitoring in securing VPN gateways.	[Apply][CO:3]
19.	Explain the different techniques used for eavesdropping attacks on VPN communications.	[Understand][CO:1,2]
20.	Differentiate between DES and AES with respect to key size, block size, security level, and performance.	[Analyze][CO:2,3]

[5x6 = 30]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

21.	Compare the working of IKEv1 and IKEv2 with respect to their secure key establishment process**.**	[Analyze] [CO:2,3]
22.	Illustrate the difference between RBAC and traditional user-based access control used in VPN user management with a simple example.	[Apply][CO:3]
23.	Describe the key security considerations of a VPN gateway, including authentication, encryption, firewall, and logging.	[Analyze] [CO:2]

[10x2 = 20]